

Total No. of Questions : 8]

SEAT No. :

PC-2673

[Total No. of Pages : 2

[6354] - 839

B.E. (Computer Engineering)

(Honours* in Artificial Intelligence and Machine Learning)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (410301)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Make suitable assumption whenever necessary.

Q1) a) Explain support vector machine. What is the difference in idea for using support vector machine for regression and classification? [8]

b) Define random forest. Explain the working of random forest algorithm in detail. [9]

OR

Q2) a) Describe a situation when you would prefer using SVM over a neural network and why? [8]

b) Explain the following with respect to random forest. [9]

- i) Bias
- ii) Variance

Q3) a) Explain Backpropagation Algorithm with respect multilayer perceptron. [9]

b) How to train perceptron? State the reasons while perceptron training samples are given one by one instead of Whole. [9]

P.T.O.

OR

- Q4)** a) Explain training procedures in multilayer perceptron in detail. [9]
b) Explain Bayesian networks in detail with suitable example. [9]
- Q5)** a) Explain the differences between supervised and unsupervised learning in detail. [8]
b) Explain K means clustering algorithm in detail. [9]

OR

- Q6)** a) Write a short notes on [8]
i) Association Rules
ii) PCA - Spectral clustering
b) Explain Apriori Algorithm with an example in detail. [9]
- Q7)** a) Formalize a second - order Markov model. What are the parameters? How can we calculate the probability of a given state sequence? How can the parameters be learned for the case of an observable model? [9]
b) Explain graphical models with respect to Hidden Markov models. [9]

OR

- Q8)** a) Write a short notes on [9]
i) Junction trees
ii) Discrete Markov process
b) State Bayes theorem. Elaborate Naive Bayes classifier with an example. [9]

